Numerical Methods

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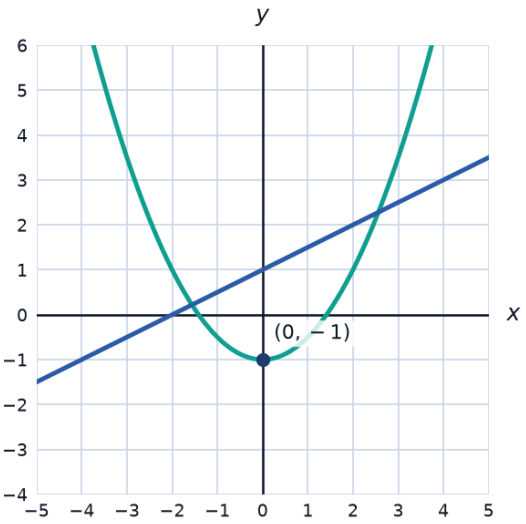
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = x^3 - 4x - 2$ has a root between $x = 2$ and $x = 3$.

2 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = x^3 - x - 1$ has a root between $x = 1$ and $x = 2$.



3 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = \ln x - 2 + x$ has a root between $x = 1$ and $x = 2$.

4 [Foundational] • Target: Locating roots by a change of sign

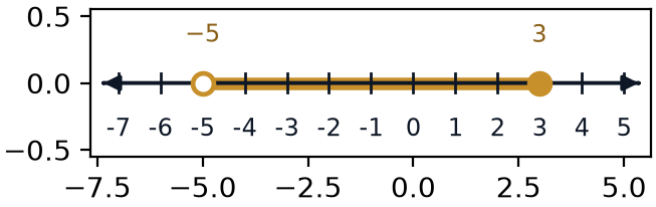
Show that $f(x) = x^3 - 2x - 5$ has a root between $x = 2$ and $x = 3$.

5 [Foundational] • Target: Interval bisection

Using $f(x) = x^3 - x - 1$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval containing the root.

6 [Foundational] • Target: Interval bisection

Continuing from the previous problem, evaluate $f(1.25)$ and give a narrower interval for the root.



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7 [Foundational] • Target: Interval bisection

Using $f(x) = \ln x - 2 + x$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval for the root.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Fixed-point iteration

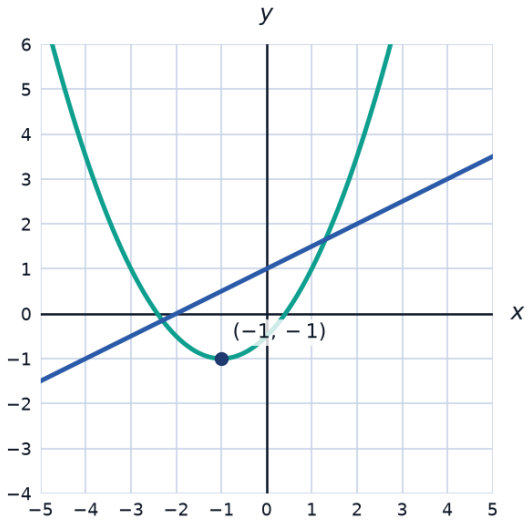
The equation $x^3 - 4x - 2 = 0$ rearranges to $x = \sqrt[3]{4x + 2}$. Using $x_0 = 2$, find x_1, x_2, and x_3, to four decimal places.

9 [Intermediate] • Target: Fixed-point iteration

The equation $x^2 - 5x + 2 = 0$ rearranges to $x = 5 - \frac{2}{x}$. Using $x_0 = 4$, find x_1 and x_2, to four decimal places.

10 [Intermediate] • Target: Fixed-point iteration

The equation $x^3 - 2x - 5 = 0$ rearranges to $x = \sqrt[3]{2x + 5}$. Using $x_0 = 2$, find x_1, to four decimal places.



11 [Intermediate] • Target: Convergence and divergence of iterations

Explain why the iteration $x_{n+1} = x_n^3 - 3$ (i.e. $g(x) = x^3 - 3$), starting from $x_0 = 1.5$, would fail to converge to a root of $x^3 - x - 3 = 0$ near $x = 1.5$.

12 [Intermediate] • Target: Convergence and divergence of iterations

Using $g(x) = 5 - \frac{2}{x}$ from the earlier iteration, find $g'(x)$ and evaluate $|g'(x)|$ at $x = 4.56$ (near the root), to explain why the iteration converges there.

13 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^2 - 3x - 1 = 0$ into the form $x = g(x)$ by making the linear term the subject.

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14 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^3 + x - 1 = 0$ into the form $x = \frac{1}{x^2 + 1}$, and verify algebraically that this is equivalent to the original equation.

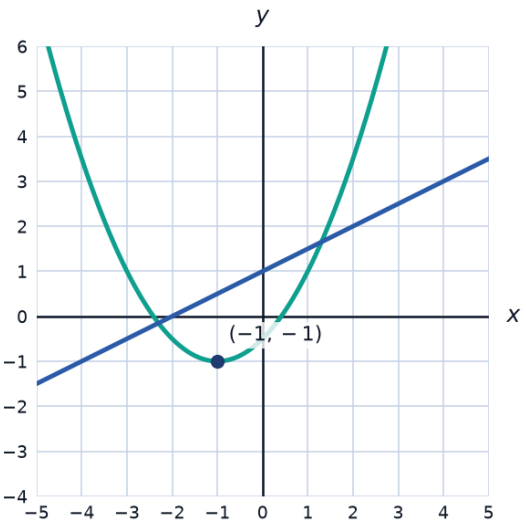
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 11.01 Differential Equations

Use one iteration of the Newton-Raphson method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 4x - 2$, to four decimal places.

16 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 12.01 Poisson Distribution

Continuing from $x_1 = 2.25$ in the previous problem, find x_2, to four decimal places.



17 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 13.01 Continuous Random Variables

Use the Newton-Raphson method with $x_0 = 2.5$ to find x_1 for $f(x) = x^2 - 7$ (approximating $\sqrt{7}$), to four decimal places.

18 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 14.01 Sampling & Estimation

Use the Newton-Raphson method with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 2x - 5$, to four decimal places.

19 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 15.01 Hypothesis Testing

Using $x_1 = 2.65$ from the previous problem for $f(x) = x^2 - 7$, perform a second iteration to find x_2, to four decimal places, and compare with $\sqrt{7} \approx 2.6458$.

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20 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 2.01 Exponentials & Logarithms

Explain why $x_0 = 1$ is an unsuitable starting point for the Newton-Raphson method applied to $f(x) = x^3 - 3x + 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Location of Roots & Sign Change

Show that the equation $x^3 - 2x - 5 = 0$ has a real root in the interval $[2, 3]$.

a. $f(2) = -1 < 0$ and $f(3) = 16 > 0$, so by IVT a root exists

b. $f(2) = 1 > 0$ and $f(3) = -2 < 0$

c. $f'(x) > 0$ for all x

d. $f(2.5) = 0$

22 [Foundational] • Target: Location of Roots & Sign Change

Under which condition does a sign change of $f(x)$ between a and b fail to guarantee a root in (a, b)?

a. When $f(x)$ is not continuous on $[a, b]$

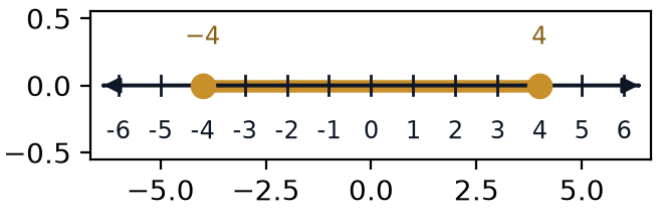
b. When $f'(x)$ changes sign

c. When $f(x)$ has an inflection point

d. When the interval is wider than 1 unit

23 [Foundational] • Target: Location of Roots & Sign Change

Use the sign change method to determine which interval of length 0.1 contains the root of $x^3 = 20$.



a. [2.7, 2.8]

b. [2.5, 2.6]

c. [2.6, 2.7]

d. [2.8, 2.9]

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24 [Foundational] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$

Rearrange the equation $x^3 - x - 1 = 0$ into an iterative formula $x_{n+1} = g(x_n)$.

a. $x_{n+1} = \sqrt[3]{x_n + 1}$

b. $x_{n+1} = x_n^3 - 1$

c. $x_{n+1} = \frac{1}{x_n^2 - 1}$

d. $x_{n+1} = x_n^3 + x_n$

25 [Foundational] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$

State the condition on $g'(x)$ for the iteration $x_{n+1} = g(x_n)$ to converge to a root α.

a. $|g'(\alpha)| < 1$

b. $|g'(\alpha)| > 1$

c. $g'(\alpha) = 0$

d. $g'(\alpha) > 0$

26 [Foundational] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$

Starting with $x_1 = 1.3$, use $x_{n+1} = \sqrt[3]{x_n + 1}$ to calculate x_2 and x_3 correct to 3 decimal places.

a. $x_2 = 1.320, x_3 = 1.324$

b. $x_2 = 1.300, x_3 = 1.310$

c. $x_2 = 1.350, x_3 = 1.380$

d. $x_2 = 1.315, x_3 = 1.330$

27 [Foundational] • Target: Cobweb and Staircase Diagrams

If $0 < g'(\alpha) < 1$ at the root, what pattern is produced on the iteration diagram?

a. A staircase diagram converging to the root

b. A cobweb diagram converging to the root

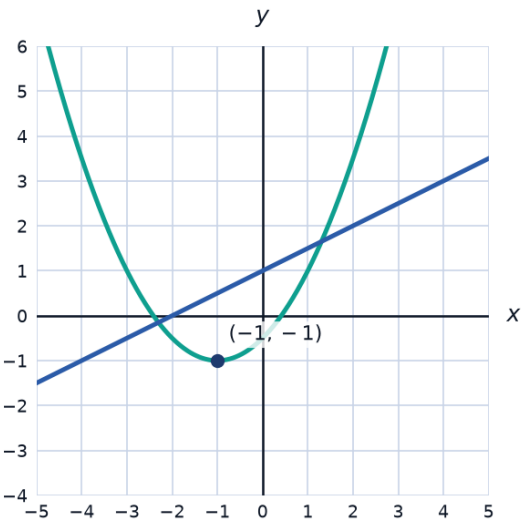
c. A diverging staircase diagram

d. An oscillating cycle

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28 [Foundational] • Target: Cobweb and Staircase Diagrams

If $-1 < g'(\alpha) < 0$ at the root, what pattern is produced on the iteration diagram?



a. A cobweb diagram spiraling inwards (converging)

b. A staircase diagram spiraling outwards

c. A cobweb diagram spiraling outwards (diverging)

d. A vertical asymptote

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cobweb and Staircase Diagrams

What happens when $|g'(\alpha)| > 1$?

a. The iteration diverges away from the root

b. The iteration converges very quickly

c. The iteration oscillates between two fixed values

d. The root becomes imaginary

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30 [Intermediate] • Target: Newton-Raphson & Advanced Applications

State the Newton-Raphson iteration formula for finding roots of $f(x) = 0$.

a. $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

b. $x_{n+1} = x_n + \frac{f(x_n)}{f'(x_n)}$

c. $x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$

d. $x_{n+1} = \frac{x_n f'(x_n) - f(x_n)}{2}$

31 [Intermediate] • Target: Newton-Raphson & Advanced Applications

Use one iteration of Newton-Raphson with $x_0 = 2$ to approximate the root of $f(x) = x^2 - 5$.

a. $x_1 = 2.25$

b. $x_1 = 2.20$

c. $x_1 = 2.30$

d. $x_1 = 2.15$

32 [Intermediate] • Target: Newton-Raphson & Advanced Applications

Under which circumstance will the Newton-Raphson method fail completely at an iteration step x_n?

a. When $f'(x_n) = 0$ (tangent is horizontal)

b. When $f(x_n) = 0$

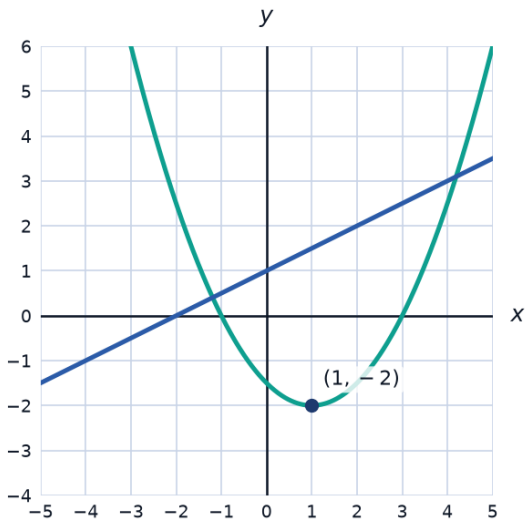
c. When $f''(x_n) = 0$

d. When x_n is negative

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33 [Intermediate] • Target: Newton-Raphson & Advanced Applications

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.



a. $k = 5, \quad x_2 = \frac{28}{9}$

b. $k = 5, \quad x_2 = \frac{44}{9}$

c. $k = 2, \quad x_2 = \frac{14}{5}$

d. $k = 7, \quad x_2 = \frac{84}{25}$

34 [Intermediate] • Target: Locating roots by a change of sign

Using the Newton-Raphson method with $f(x) = x^3 - 53$ and $x_0 = -8$, find x_1.

a. $-\frac{779}{192}$

b. $\frac{971}{192}$

c. $-\frac{971}{192}$

d. -8

35 [Intermediate] • Target: Interval bisection

Using the Newton-Raphson method with $f(x) = x^3 - 48$ and $x_0 = 7$, find x_1.

a. 7

b. $-\frac{734}{147}$

c. $\frac{881}{147}$

d. $\frac{734}{147}$

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36 [Intermediate] • Target: Fixed-point iteration

Using the Newton-Raphson method with $f(x) = x^3 - 10$ and $x_0 = 3$, find x_1.

a. $\frac{91}{27}$

b. 3

c. $-\frac{64}{27}$

d. $\frac{64}{27}$

37 [Intermediate] • Target: Convergence and divergence of iterations

Using the Newton-Raphson method with $f(x) = x^3 - 33$ and $x_0 = -3$, find x_1.

a. $-\frac{7}{9}$

b. $\frac{7}{9}$

c. $\frac{2}{9}$

d. -3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rearranging equations for iteration • Cross-Lesson: 14.01 Sampling & Estimation

Using the Newton-Raphson method with $f(x) = x^3 - 2$ and $x_0 = 4$, find x_1.

a. $\frac{65}{24}$

b. $-\frac{65}{24}$

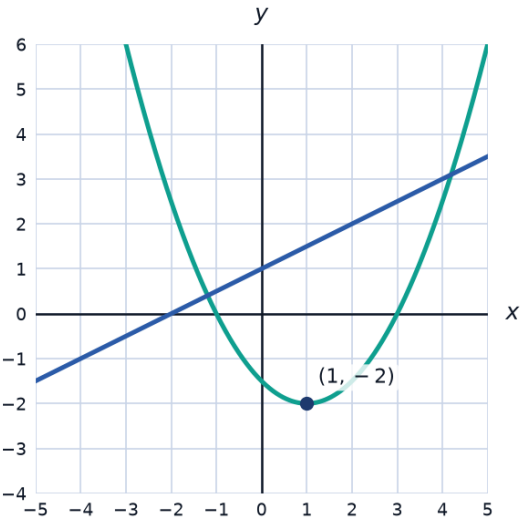
c. 4

d. $\frac{89}{24}$

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39 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 15.01 Hypothesis Testing

Using the Newton-Raphson method with $f(x) = x^3 - 56$ and $x_0 = 1$, find x_1.



a. 1

b. $\frac{61}{3}$

c. $-\frac{58}{3}$

d. $\frac{58}{3}$

40 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 2.01 Exponentials & Logarithms

Using the Newton-Raphson method with $f(x) = x^2 - 18$ and $x_0 = 5$, find x_1.

a. 5

b. $\frac{43}{10}$

c. $-\frac{43}{10}$

d. $\frac{53}{10}$

41 [Challenging] • Target: Error bounds from bisection • Cross-Lesson: 3.01 Advanced Trigonometry

Using the Newton-Raphson method with $f(x) = x^2 - 14$ and $x_0 = 5$, find x_1.

a. $\frac{49}{10}$

b. $-\frac{39}{10}$

c. 5

d. $\frac{39}{10}$

42 [Challenging] • Target: A well-designed word problem: designing a support strut • Cross-Lesson: 4.01 Differentiation

Using the Newton-Raphson method with $f(x) = x^2 - 13$ and $x_0 = 6$, find x_1.

a. 6

b. $\frac{61}{12}$

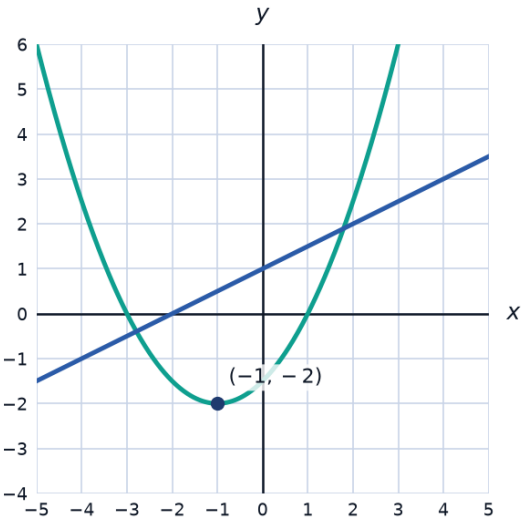
c. $\frac{49}{12}$

d. $-\frac{49}{12}$

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43 [Challenging] • Target: Location of Roots & Sign Change • Cross-Lesson: 5.01 Integration

Using the Newton-Raphson method with $f(x) = x^2 - 22$ and $x_0 = -3$, find x_1.



a. $-\frac{25}{6}$

b. $-\frac{31}{6}$

c. $\frac{31}{6}$

d. -3

44 [Challenging] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$ • Cross-Lesson: 6.01 Sequences & Series

Using the Newton-Raphson method with $f(x) = x^2 - 13$ and $x_0 = 2$, find x_1.

a. 2

b. $-\frac{17}{4}$

c. $\frac{21}{4}$

d. $\frac{17}{4}$

45 [Challenging] • Target: Cobweb and Staircase Diagrams • Cross-Lesson: 7.01 Vectors

Using the Newton-Raphson method with $f(x) = x^2 - 18$ and $x_0 = 4$, find x_1.

a. 4

b. $-\frac{17}{4}$

c. $\frac{21}{4}$

d. $\frac{17}{4}$

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